

g2(0) cases

I.V. Krainov^{1,*} and A.M. Kononov¹

¹*Ioffe Institute, 194021 St. Petersburg, Russia*

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abstract

In this paper, we consider a charged quantum dot (QD) with resident electron located in optical microresonator. Optically active transition is transition between ground state and charged trion with energy E_t , equaling to energy of the fundamental resonator mode $\hbar\omega_0$. The sistem is excited by laser pump with detuning Δ . Hameltonian of that sistem in the rotation wave approximation is:

$$\hat{H} = E_t \left(\hat{d}_\uparrow^\dagger \hat{d}_\uparrow + \hat{d}_\downarrow^\dagger \hat{d}_\downarrow \right) + \omega_0 \left(\hat{b}_+^\dagger \hat{b}_+ + \hat{b}_-^\dagger \hat{b}_- \right) + \quad (1)$$

$$+ g \left(\hat{d}_\uparrow^\dagger \hat{b}_+ \hat{a}_\uparrow + \hat{d}_\downarrow^\dagger \hat{b}_- \hat{a}_\downarrow \right) + g^* \left(\hat{d}_\uparrow \hat{b}_+^\dagger \hat{a}_\uparrow^\dagger + \hat{d}_\downarrow \hat{b}_-^\dagger \hat{a}_\downarrow^\dagger \right)$$

since the pumping of the system is carried out by a laser pulse with a large number of photons, the interaction with this part can be described by a classical field:

$$\hat{H} = \Delta \hat{d}_\uparrow^\dagger \hat{d}_\uparrow + \Delta \hat{b}_+^\dagger \hat{b}_+ + \Omega_R \left(\hat{d}_\uparrow^\dagger \hat{a}_\uparrow + \hat{d}_\uparrow \hat{a}_\uparrow^\dagger \right) + g \left(\hat{b}_+ \hat{d}_\uparrow^\dagger \hat{a}_\uparrow + \hat{b}_+^\dagger \hat{d}_\uparrow \hat{a}_\uparrow^\dagger \right) - \frac{i}{\hbar} \left[\hat{H}; \hat{\rho} \right] + \sum_i \gamma_i \left(\hat{V}_i^\dagger \hat{V}_i \hat{\rho} + \hat{\rho} \hat{V}_i^\dagger \hat{V}_i - 2 \hat{V}_i \hat{\rho} \hat{V}_i^\dagger \right) \equiv \hat{\mathcal{L}}t \quad (2)$$

$$+ \Delta \hat{d}_\downarrow^\dagger \hat{d}_\downarrow + \Delta \hat{b}_-^\dagger \hat{b}_- + \Omega_R \left(\hat{d}_\downarrow^\dagger \hat{a}_\downarrow + \hat{d}_\downarrow \hat{a}_\downarrow^\dagger \right) + g \left(\hat{b}_- \hat{d}_\downarrow^\dagger \hat{a}_\downarrow + \hat{b}_-^\dagger \hat{d}_\downarrow \hat{a}_\downarrow^\dagger \right) \quad (4)$$

where the optical mode creation operator in circular polarisation $\sigma_\pm$: $\hat{b}_\pm^\dagger = (\hat{b}_+^\dagger \pm \hat{b}_-^\dagger) / \sqrt{2}$; the electron creation

operator $\hat{a}_{\uparrow\downarrow}^\dagger$, where $\uparrow\downarrow$ denotes the electron state $|\pm 1/2\rangle$; the trion creation operator $\hat{d}_{\uparrow\downarrow}^\dagger$, where $\uparrow\downarrow$ denotes the negatively charged trion state with heavy hole spin $|\pm 3/2\rangle$; g is the effective QD-cavity coupling parameter; Ω_R is Rabi frequency.

QD is located in horizontal magnetic field for controlling the spin polarisation of resident electron. Hameltonian of that intraction is:

$$\hat{H}_B = \mu_B g_e B_x \left(\hat{a}_\downarrow^\dagger \hat{a}_\uparrow + \hat{a}_\uparrow^\dagger \hat{a}_\downarrow \right) + \mu_B g_t B_x \left(\hat{d}_\downarrow^\dagger \hat{d}_\uparrow + \hat{d}_\uparrow^\dagger \hat{d}_\downarrow \right) \quad (3)$$

where g_e g-factor of electron, g_t g-factor of trion.

In order to describe evolution with the escape of photons into the environment, evolution equation have to be:

where γ_i and $\hat{V}_i$ are the rate and jump operator in the Lindbladian. It includes the photon yield from the cavity mode with $\gamma_{photon} = \Gamma$, and $\hat{V}_{photon} = \hat{b}_\pm$ and phonon pure-dephasing mechanism with $\gamma_{phonon} = \gamma \Omega_R^2$ and $\hat{V}_{phonon} = \dots$, and

* igor.krainov@mail.ru